Def. Given a space X, a covering space of X consists of a space $\tilde{X}$ and a map $p : \tilde{X} \to X$ s.t.

For each $x \in X$, there is an open neighborhood U of x in X satisfying

$$p^{-1}[U] = \coprod_{i \in \Lambda} \mathcal{O}_i \quad \text{and} \quad p|_{\mathcal{O}_i} : \mathcal{O}_i \xrightarrow{\simeq} U_i.$$

(Such U is called "evenly covered")

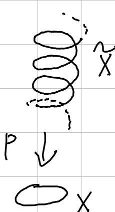
Example. Given a space S^1, let $p : \mathbb{R} \to S^1 ; s \mapsto (\cos 2\pi s, \sin 2\pi s)$.

Let $(1,0) \in S^1$ be given. Then, for a neighborhood $B_1(1,0) \cap S^1$

$$p^{-1}[B_1(1,0) \cap S^1] = \bigsqcup_{n \in \mathbb{Z}} \left(n + \left(-\tfrac{1}{6}, \tfrac{1}{6}\right)\right)$$

and for each $n \in \mathbb{Z}$,

$$\left(n + \left(-\tfrac{1}{6}, \tfrac{1}{6}\right)\right) \cong B_1(1,0) \cap S^1 \text{ via } p.$$



Def. Given a covering $p : \tilde{X} \to X$ and a map $f : Y \to X$,
a continuous map $\tilde{f} : Y \to \tilde{X}$ such that $p \circ \tilde{f} = f$ is called a lift of f to $\tilde{X}$, with respect to p.

$$\begin{tikzcd} \tilde{X} \arrow[dr, "p"] & \\ Y \arrow[u, "\tilde{f}"] \arrow[r, "f"'] & X \end{tikzcd}$$

Thm. Given a map $F: Y \times I \to X$ and a map $\tilde{F}: Y \times \{0\} \to \tilde{X}$ lifting $F|_{Y \times \{0\}}$, there is a unique map $\tilde{F}: Y \times I \to \tilde{X}$ lifting F extended $\tilde{F}: Y \times \{0\} \to \tilde{X}$.

Proof.